

Yufeng (Sylvan) Yu

Graphics / Engine Engineer · AI Engineering

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Summary

Graphics / engine engineer focused on productizing low-level rendering — pipelines, cross-platform consistency, editor toolchains, and AI-agent infrastructure — for products at 100M+ DAU scale. Currently own the rendering core and AI toolchain of **Predy**, Xiaohongshu (RedNote)'s in-house real-time motion engine; previously co-built the Galacean engine and shipped graphics for Alipay's flagship CNY campaigns at Ant Group.

Skills

- **Graphics / engine:** WebGL, GLSL shaders, render pipelines, particle systems, depth / stencil, performance & GPU-memory optimization
- **Cross-platform:** React Native, JSI, iOS / Android native integration, text layout, cross-platform consistency & regression testing
- **AI / agents:** editor agents, local MCP, custom tools, Langfuse observability, Python DCC pipelines

Experience

Xiaohongshu (RedNote) · Commerce Technology · PE Infra | Product Engineer, Graphics / Engine Jul 2025 – Present

- **TextLine cross-platform text system:** designed from scratch a single-line rich-text system unifying text metrics across **Web / RN / iOS / Android / HarmonyOS** under one protocol; native-layer text measurement exposed to JS via **JSI**; emoji atlas + baseline management; JSONL / e2e snapshot regression with hundreds of native bitmap samples — cross-platform text rendering made reproducible and regression-testable.
- **Marketing & CNY at scale:** built a fully parameterized coupon component; consolidating dynamic text into TextLine cut first frame **1.41s → 850ms (Android), 500ms → 360ms (iOS)**; **30M PV in February, >99% success rate** on both platforms; corresponding business-wide claim rate **+2.6%**, redemption **+1.6%**, DAB **+0.53%**. CNY fireworks: **100M+ plays (~10x YoY)**, 16 large scenes at **56+ FPS**, record-low crash rates (**Android <0.01%**, **iOS <0.001%**).
- **Engine internals & quality:** consolidated render modules into a standard shader / UBO system; binary-packed shape geometry (**~50% smaller**); TypedArray reuse and zero-copy uploads against GC jitter; fixed a driver-order-dependent **UBO alignment bug**; automatic WebGL context-loss recovery; shipped an After-Effects-Graph-Editor-class animation curve editor and a **pre-publish performance gate** with device-calibrated thresholds.
- **AI infrastructure:** designed a **local MCP** architecture (stdio + localhost WebSocket; avoids multi-user crosstalk by design); LangChain → **Langfuse** observability separating real errors from user cancellations; token / quota governance; one-command onboarding tool for non-engineers.
- **2.5D parallax photos:** converted single 2D images into multi-layer RGBD / parallax texture assets (Depth Anything V2 / SAM2 / LaMa) for splash-ad scenarios; ran the full **WebGL1/2 + Metal + React Native** rendering path with 0–3-layer degradation.

Ant Group · Alipay · Experience Technology | Front-End Engineer, Graphics / Engine Jul 2022 – Jul 2025

Co-built with the Galacean engine team. Awards: **Outstanding New Hire** (department & Ant Beijing).

- **Shader & rendering:** independently designed and standardized the **Uber VFX / Uber Standard shader framework** (PBR / NPR), enabling zero-code delivery for **90%+** of art requirements; shipped frustum culling, instancing, billboard static batching, and multi-pass post-processing under national-app WebGL constraints; guest speaker at a Galacean public session.
- **Complex 3D performance:** built two vegetation systems (Compute Shader for native, GPU instancing for WebGL), cutting rendering cost to **~15% of the third-party baseline**; batches / SetPass calls **–50% / –60%**; stable **40 FPS** at max quality on Snapdragon 865.
- **High-end rendering:** led top-spec in-app 3D scenes — subsurface scattering, water, vegetation, terrain blending, and mecha shaders (~10 modules) — with a custom HDR post pipeline and dynamic degradation strategy, adopted across multiple product lines.
- **100M-DAU interaction lead:** graphics / interaction lead on Alipay's CNY "Wufu" campaigns — Rabbit-year mini-games with pre-generated audio-spectrum textures bypassing platform API limits; Tiger-year (as intern) single-player / multiplayer / voice WebGL versions with **near-100% device coverage and 0.05% crash rate**; led Jingtan collectibles' engine migration to Galacean, decoupling 58 animation modules for **100M-scale NFT assets**.
- **Toolchain, digital humans & XR:** Unity–Galacean art-workflow tooling; digital-human shaders powering the **Hangzhou Asian Games digital torchbearer**; Python Maya / Blender pipelines for AI 3D dataset standardization; **early individual Apple Vision Pro developer** — shipped a 60-second MR expo game in three weeks.

Education & Earlier

Zhejiang Gongshang University — B.Eng., Electronic Information Engineering (2018 – 2022), Hangzhou; personal project published as a teaching case in a WeChat Mini-Program development textbook

Internships: Ant Group · Alipay (2021 – 2022) · ByteDance, TikTok advertiser creative platform (2021) · Hikvision (2020 – 2021)